



Exploratory Data Analysis Report

Cleaned & Feature-Engineered Dataset · Insight-Driven Analysis



1,000 Users



14 Base Features



8 Engineered Features



No Missing Values



No Outliers Detected

1,000

TOTAL USERS

41 yrs

AVG AGE

\$65.7K

AVG INCOME

\$2,526

AVG TOTAL SPEND

297 min

AVG TIME ON SITE

12.8%

POWER USERS

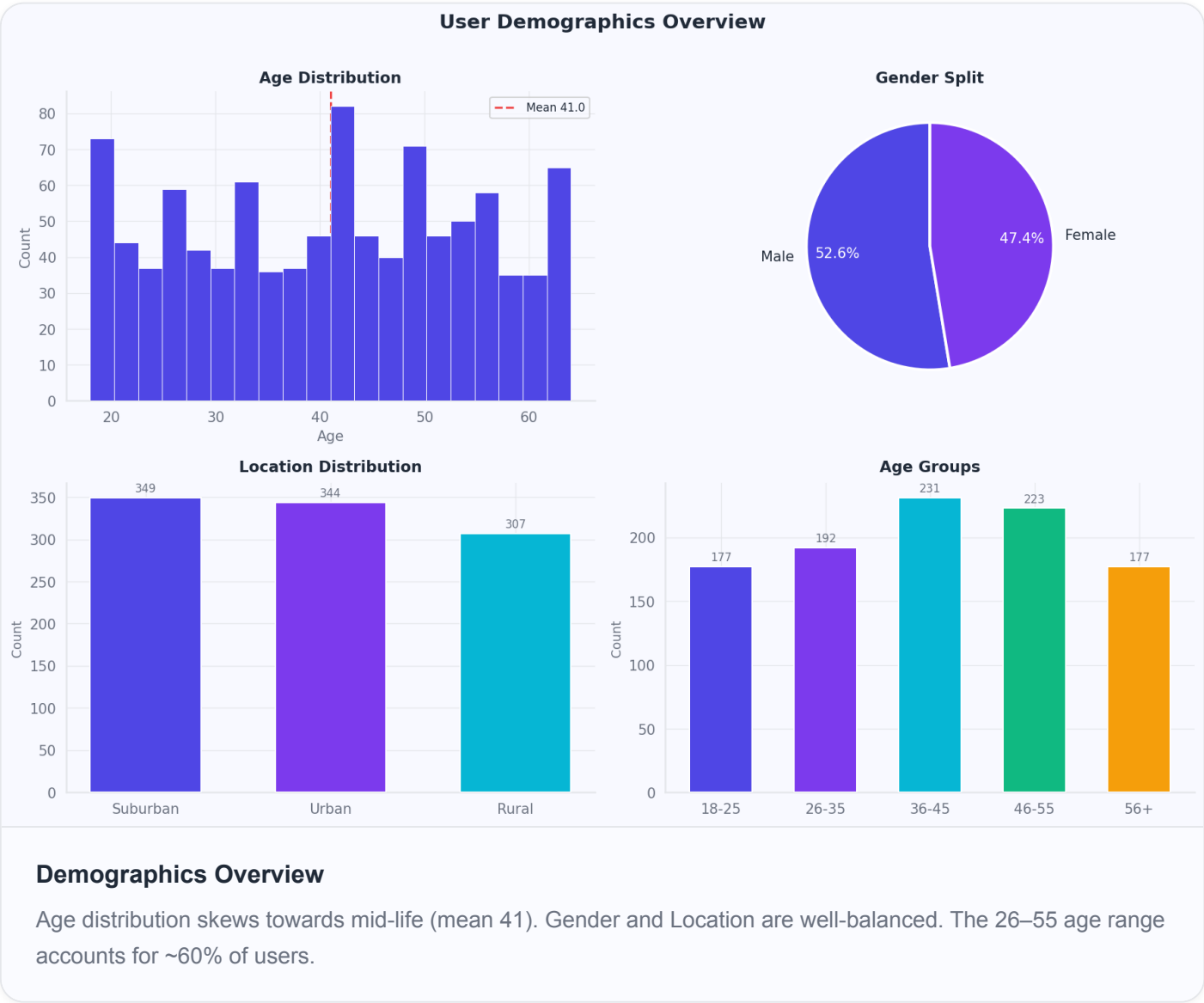
Engineered Features

8 new features derived from base columns to enable richer modelling and segmentation.

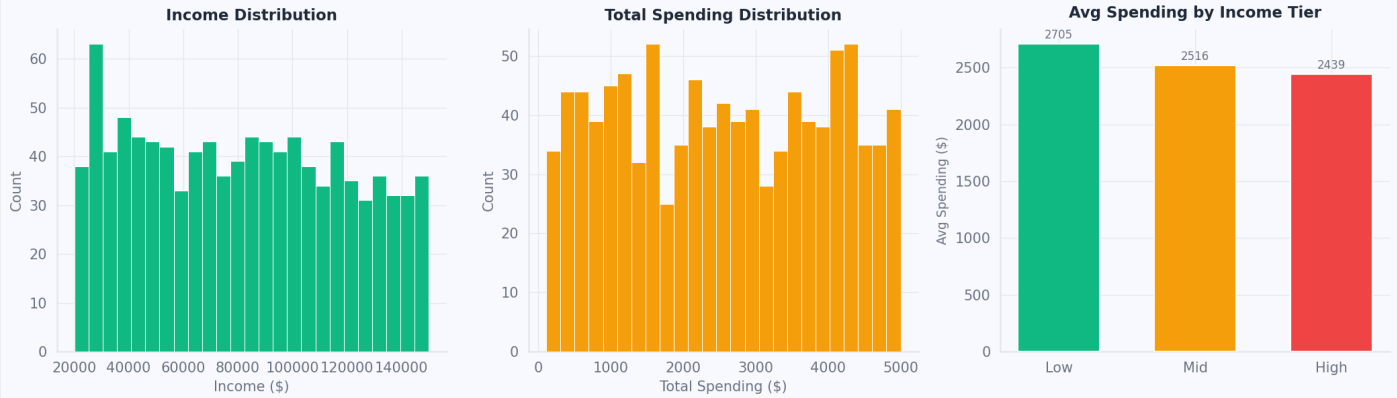
Feature	Type	Formula / Logic	Purpose
Age_Group	Ordinal Cat.	pd.cut(Age, [0,25,35,45,55,100])	Life-stage segmentation
Income_Tier	Ordinal Cat.	Tertile split (33rd / 67th pct)	Affordability segment
Engagement_Score	Float [0–1]	$0.35 \times \text{Time} + 0.30 \times \text{Pages} + 0.35 \times \text{PurchaseFreq (normalised)}$	Single engagement KPI
Revenue_Per_Visit	Float	$\text{Total_Spending} / (\text{Time_on_Site_hrs})$	Spend efficiency
Recency_Score	Int	$\text{max}(\text{Last_Login}) - \text{Last_Login}$	RFM recency component

Feature	Type	Formula / Logic	Purpose
CLV_Proxy	Int	AOV × Purchase_Frequency	Customer lifetime value proxy
Interest_Category_Match	Binary	Map Interest→expected Category, compare	Preference alignment flag
Power_User	Binary	Engagement_Score ≥ Q75 AND Newsletter=True	High-value user flag

1. Demographics Overview



Spending & Income Analysis



Spending & Income Analysis

Income is broadly uniform. Total Spending is right-skewed with most users below \$3K. High-income tier users spend only marginally more on average.

3. User Engagement Metrics

User Engagement Metrics

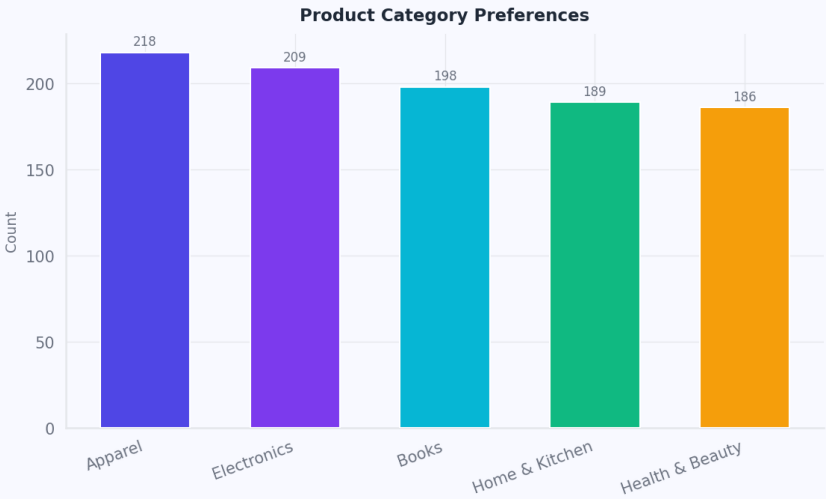
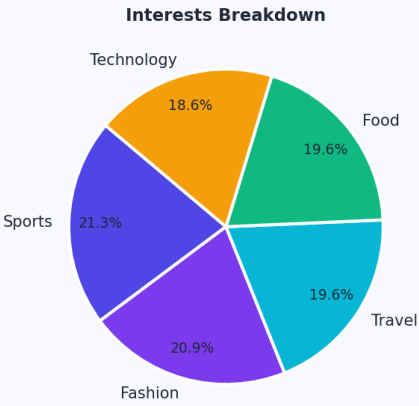


User Engagement Metrics

Engagement Score follows a roughly normal distribution centred at 0.5. Time on Site and Pages Viewed are positively correlated. Middle age groups show peak engagement.

4. Interests & Product Preferences

Interests & Product Preferences

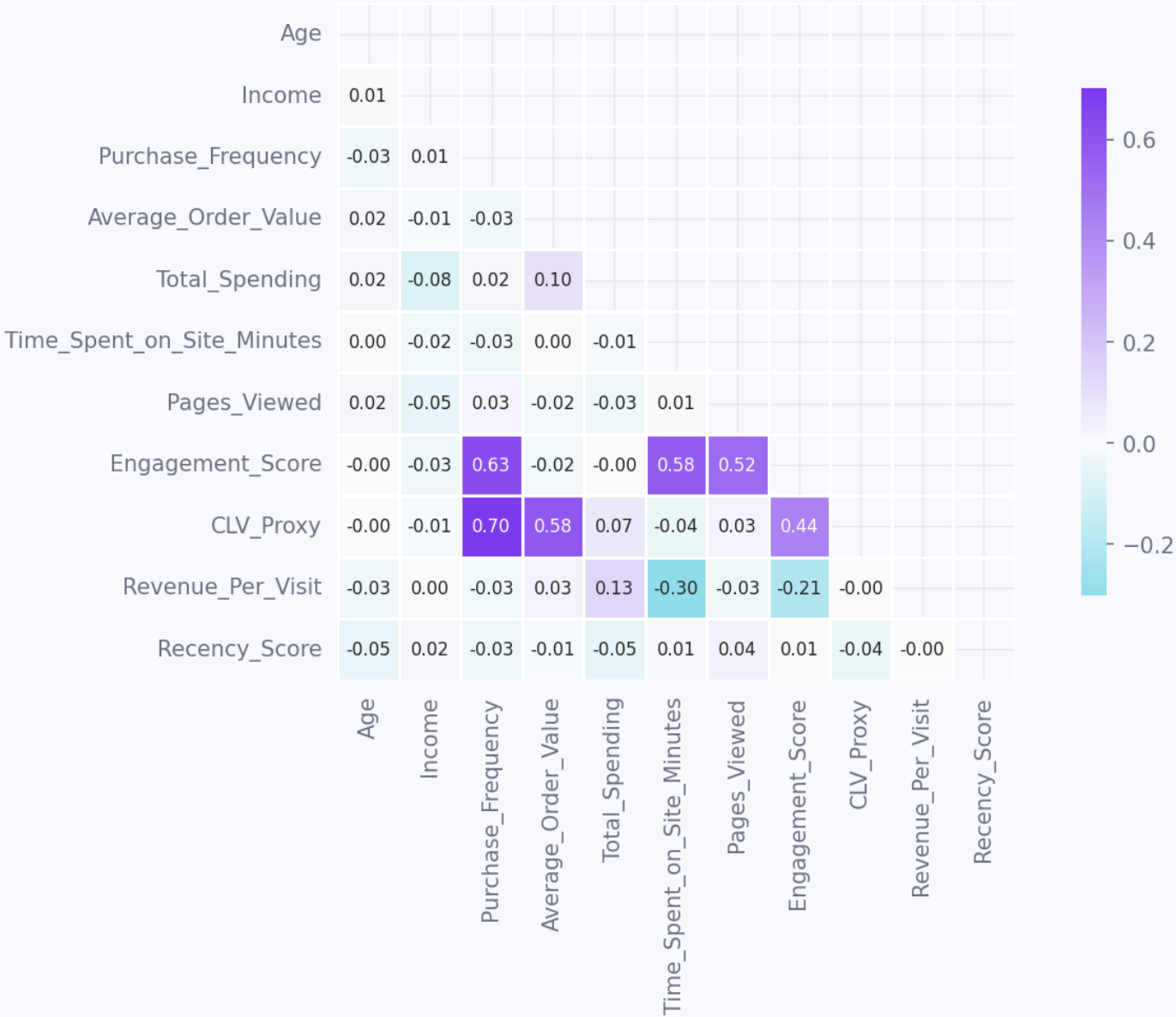


Interests & Product Preferences

Interests and preferred categories are evenly distributed across the 5 options each — no single dominant group. Apparel leads product preference by a slim margin.

5. Feature Correlation Heatmap

Feature Correlation Heatmap

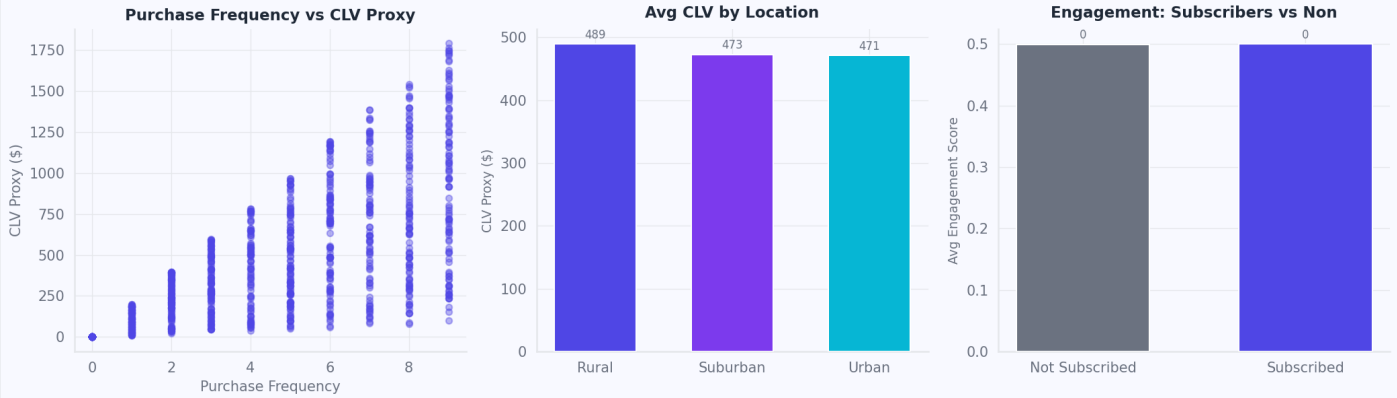


Feature Correlation Heatmap

CLV Proxy and Purchase Frequency are highly correlated. Engagement Score correlates moderately with Pages Viewed and Time on Site. Income shows near-zero correlation with most behavioural metrics.

6. Revenue & CLV Insights

Revenue & CLV Insights

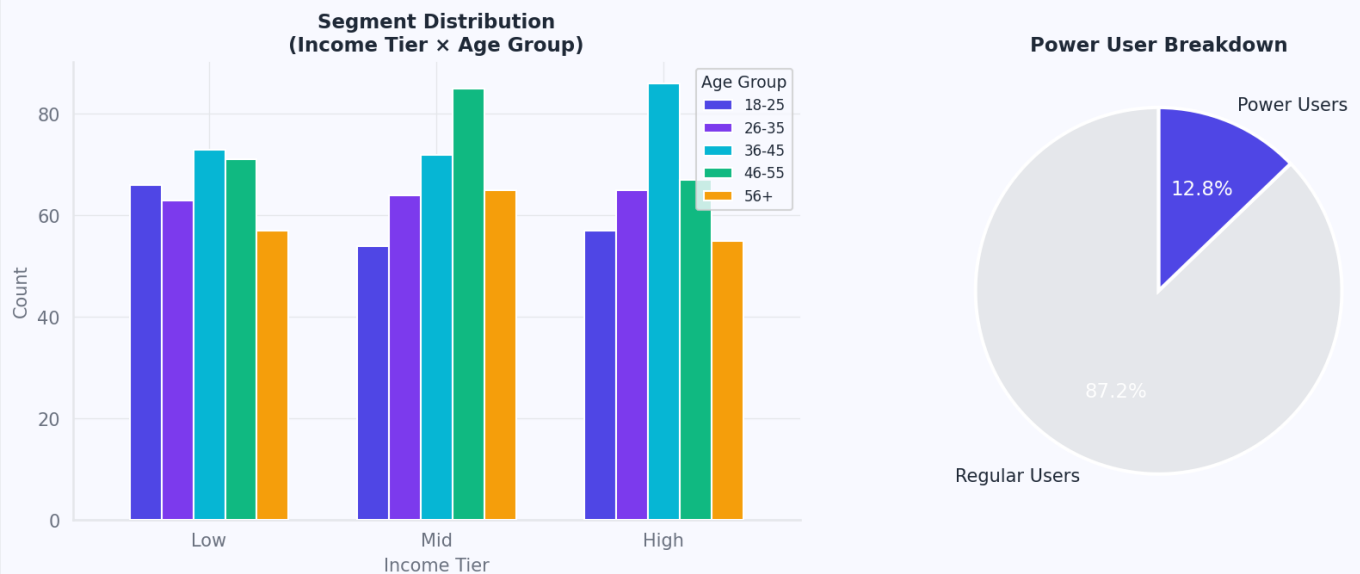


Revenue & CLV Insights

CLV Proxy scales linearly with Purchase Frequency. Location has minimal influence on CLV. Newsletter subscribers show a clear engagement advantage.

7. User Segmentation & Power Users

User Segmentation & Power Users



User Segmentation & Power Users

Segment sizes are balanced across Income Tier x Age Group combinations. Power Users (12.8%) are the top-engagement, newsletter-subscribed cohort worth targeting for retention campaigns.